In summary, Auto Scaling groups are a useful tool if you need to start multiple virtual machines of the same kind across multiple availability zones. Additionally, an Auto Scaling group replaces failed EC2 instances automatically.

17.2 Using metrics or schedules to trigger scaling

So far in this chapter, you've learned how to use an Auto Scaling group and a launch template to manage virtual machines. With that in mind, you can change the desired capacity of the Auto Scaling group manually so new instances will be started or old instances will be terminated to reach the new desired capacity.

To provide a scalable infrastructure for a blogging platform, you need to increase and decrease the number of virtual machines in the pool automatically by adjusting the desired capacity of the Auto Scaling group with scaling policies. Many people surf the web during their lunch break, so you might need to add virtual machines every day between 11 a.m. and 1 p.m. You also need to adapt to unpredictable load patterns—for example, if articles hosted on your blogging platform are shared frequently through social networks. Figure 17.3 illustrates two ways to change the number of virtual machines, as described in the following list.

- *Defining a schedule*—The timing would increase or decrease the number of virtual machines according to recurring load patterns (such as decreasing the number of virtual machines at night).
- *Using a CloudWatch alarm*—The alarm will trigger a scaling policy to increase or decrease the number of virtual machines based on a metric (such as CPU usage or number of requests on the load balancer).